THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The number of perfect squares between 200 and 900 (both excluded) is: 1

(a) 15 **(b)** 17 **(c)** 14 **(d)** 16

2. 2^7 added 2 times equals: 1

(a) 2^8 **(b)** 4^7 **(c)** 2^{14} **(d)** 2^9

3. If $(34)_b = 19$ in decimal (a two-digit number in base b), then b equals: 1

(a) 4 **(b)** 6 **(c)** 7 **(d)** 5

4. The angles of a quadrilateral are in the ratio 3:4:5:8. The difference between the largest and the smallest angle is: 1

(a) 198° **(b)** 90° **(c)** 102° **(d)** 144°

5. The smallest number which leaves remainder 5 when divided by 12 and 18 is: 1

(a) 41 **(b)** 221 **(c)** 36 **(d)** 77

6. The smallest number to be added to 1234 to make it a perfect square is: 1

(a) 61 **(b)** 64 **(c)** 63 **(d)** 62

7. If $2^{2x+1} = 2^9$, the value of x is: 1

(a) 8 **(b)** 4 **(c)** 5 **(d)** 3

8. Weights of 1, 2, 4, 8, ... kg (powers of 2) are available. The least number of weights needed to weigh any whole number of kg up to 31 kg is: 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

9. One angle of a parallelogram is 2 times its adjacent angle. The smaller angle is: 1

(a) 120° **(b)** 80° **(c)** 60° **(d)** 90°

10. The number of multiples of 7 between 100 and 500 (both included) is: 1

(a) 57 **(b)** 58 **(c)** 56 **(d)** 59

11. The largest number to be subtracted from 2577 to make it a perfect square is: 1

(a) 80 **(b)** 77 **(c)** 79 **(d)** 78

12. The number of digits in $2^4 \times 5^7$ is: 1

(a) 9 **(b)** 8 **(c)** 6 **(d)** 7

13. Which of the following numbers is the greatest? 1

(a) 32_5 **(b)** 14_8 **(c)** 1101_2 **(d)** 16_{10}

14. In a rhombus ABCD, $\angle ACD = 25^\circ$. Then $\angle ABC$ is: 1

(a) 50° **(b)** 155° **(c)** 130° **(d)** 65°

15. The number of 3-digit numbers divisible by 8 is: 1

(a) 124 **(b)** 113 **(c)** 112 **(d)** 111

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the smallest number to be added to 3900 to make it a perfect square. What is the square root of the result? 2

17. Find x if $3^{2x-1} = 243$. 2

18. Find the base b if $(23)_b = 13$ in decimal. 2

19. In a parallelogram ABCD, $\angle A = (4x + 14)^\circ$ and $\angle C = (5x + 2)^\circ$. Find x and $\angle B$. 2

20. Find the smallest number which leaves remainder 3 when divided by 20 and 25. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A square garden has area 1225 m². Fencing costs ₹40 per metre. Find the total cost of fencing it. 3

22. A sheet of paper 0.1 mm thick is folded in half 10 times (assume this is possible). Find the thickness of the folded paper in mm and in cm. 3

23. A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 45 g? Explain using binary numbers. 3

24. In a parallelogram ABCD with $\angle A = 50^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

OR

Show that the six-digit number 247247 is divisible by 7, 11 and 13. 3

25. A solid cube of side 6 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

OR

Light travels 3×10^8 m in one second. How far does it travel in 2 minutes? Give the answer in scientific notation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the smallest square number that is divisible by each of 6, 8 and 12. 5

OR

A culture has 5 bacteria and the number becomes 3 times every hour. Write the number after 4 hours as an expression with a power, find its value, and say how many times the original number it is. 5

27. In some base b the number $(234)_b$ equals 69 in decimal. Find b and write the number in base 10 form. 5

OR

In a quadrilateral ABCD, $\angle A : \angle B : \angle C : \angle D = 1 : 2 : 3 : 4$. Find the four angles. Show that AB $\parallel$ DC. 5

END OF QUESTION PAPER **Total: 50 Marks**

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